

Problem Set 5

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Question 1.

Code:

```
#####
# q1
# i
setwd("C:/Users/Айдос/OneDrive - nu.edu.kz/MSF/Core 9/Econometrics/ps5")
Signals = read.csv('Signals_A_62016.csv', header=TRUE, as.is=TRUE, na.strings =
c("NA", ".", ""))
Signals$PFormDate = as.Date(Signals$PFormDate, format = '%d/%m/%Y')

# ii
Signals$qNXF = ave(Signals$NXF,
                  Signals$PFormDate,
                  FUN= function(x) {cut(x, breaks=quantile(x, probs=seq(from=0,
to=1,
                                by=0.2), na.rm=TRUE), include.lowest=TRUE, na.rm=TRUE)})

# iii
SignalsDF_2002 = Signals[Signals$PFormDate == '2002-06-30',]

# iv
mean(SignalsDF_2002$AvgMonRet[SignalsDF_2002$qNXF == 1], na.rm = TRUE)
0.4453955
median(SignalsDF_2002$AvgMonRet[SignalsDF_2002$qNXF == 1], na.rm = TRUE)
0.2258
sd(SignalsDF_2002$AvgMonRet[SignalsDF_2002$qNXF == 1], na.rm = TRUE)
2.904674
skewness(SignalsDF_2002$AvgMonRet[SignalsDF_2002$qNXF == 1], na.rm = TRUE)
0.9619619
kurtosis(SignalsDF_2002$AvgMonRet[SignalsDF_2002$qNXF == 1], na.rm = TRUE)
8.887628
cor(SignalsDF_2002[c('AvgMonRet', 'Size')][SignalsDF_2002$qNXF ==
1,][,1], SignalsDF_2002[c('AvgMonRet', 'Size')][SignalsDF_2002$qNXF == 1,][,2], use =
'complete.obs', method = 'pearson')
-0.02174134

mean(SignalsDF_2002$AvgMonRet[SignalsDF_2002$qNXF == 5], na.rm = TRUE)
1.565114
median(SignalsDF_2002$AvgMonRet[SignalsDF_2002$qNXF == 5], na.rm = TRUE)
0.8431583
sd(SignalsDF_2002$AvgMonRet[SignalsDF_2002$qNXF == 5], na.rm = TRUE)
4.399157
skewness(SignalsDF_2002$AvgMonRet[SignalsDF_2002$qNXF == 5], na.rm = TRUE)
1.047373
kurtosis(SignalsDF_2002$AvgMonRet[SignalsDF_2002$qNXF == 5], na.rm = TRUE)
6.89181
cor(SignalsDF_2002[c('AvgMonRet', 'Size')][SignalsDF_2002$qNXF ==
5,][,1], SignalsDF_2002[c('AvgMonRet', 'Size')][SignalsDF_2002$qNXF == 5,][,2], use =
'complete.obs', method = 'pearson')
-0.04338349
```

We can see that both distributions have a positive skewness (left-modal), i.e. mean is greater than median value. The standard deviation for qNXF=1 portfolio is 2.9 which is less than qNXF=5 portfolio's standard deviation of 4.4. Both portfolios have a kurtosis larger than the kurtosis of the standard normal distribution. Large kurtosis is desirable for stock returns, because it means that there is less value at risk, i.e. the probabilities of critical outcomes is smaller than in SND.

In addition, in both portfolios we obtained a negative Pearson correlation between AvgMonRet and Size.

```
# v
jarque.bera.test(SignalsDF_2002$AvgMonRet[SignalsDF_2002$qNXF ==
1][!is.na(SignalsDF_2002$AvgMonRet[SignalsDF_2002$qNXF == 1])])
```

Jarque Bera Test

```
data: SignalsDF_2002$AvgMonRet[SignalsDF_2002$qNXF == 1][!is.na(SignalsDF
_2002$AvgMonRet[SignalsDF_2002$qNXF == 1])]
X-squared = 848.84, df = 2, p-value < 0.00000000000000022
```

```
jarque.bera.test(SignalsDF_2002$AvgMonRet[SignalsDF_2002$qNXF ==
5][!is.na(SignalsDF_2002$AvgMonRet[SignalsDF_2002$qNXF == 5])])
```

Jarque Bera Test

```
data: SignalsDF_2002$AvgMonRet[SignalsDF_2002$qNXF == 5][!is.na(SignalsDF
_2002$AvgMonRet[SignalsDF_2002$qNXF == 5])]
X-squared = 435.45, df = 2, p-value < 0.00000000000000022
```

In both cases we reject the normality via Jarque Bera test. The Jarque and Bera test combines estimators for skewness and kurtosis into one test statistic.

Indeed, both distributions do not follow the standard normal distribution, because both are skewed, and have a kurtosis larger than 3. Also, mean and median are not equal.

Since it fails the normality test, this implies that the OLS estimators are inconsistent (inefficient).

```
# vi
# median for qNXF = 1
X = SignalsDF_2002$AvgMonRet[SignalsDF_2002$qNXF ==
1][!is.na(SignalsDF_2002$AvgMonRet[SignalsDF_2002$qNXF == 1])]
orig_sample = X
n = length(orig_sample)

boot_samples = replicate(n=100000, sample(orig_sample, n, replace=TRUE),
simplify=FALSE)
boot_estimates = sapply(boot_samples, FUN=median, simplify=TRUE)
se_boot = sd(boot_estimates)
orig_sample_estimate = median(orig_sample)

CI_95_se_method = c(orig_sample_estimate + qnorm(p=0.025, mean=0, sd=1)*se_boot,
orig_sample_estimate - qnorm(p=0.025, mean=0, sd=1)*se_boot)
CI_95_se_method
```

```
0.06183005 0.38976995
```

```
qs_boot = quantile(boot_estimates, probs=c(0.025, 0.975))
CI_95_p_method = unname(qs_boot)
CI_95_p_method
```

```
0.09030833 0.40925000
```

```
# median for qNXF = 5
X = SignalsDF_2002$AvgMonRet[SignalsDF_2002$qNXF ==
5][!is.na(SignalsDF_2002$AvgMonRet[SignalsDF_2002$qNXF == 5])]
orig_sample = X
n = length(orig_sample)

boot_samples = replicate(n=100000, sample(orig_sample, n, replace=TRUE),
simplify=FALSE)
boot_estimates = sapply(boot_samples, FUN=median, simplify=TRUE)
se_boot = sd(boot_estimates)
orig_sample_estimate = median(orig_sample)

CI_95_se_method = c(orig_sample_estimate + qnorm(p=0.025, mean=0, sd=1)*se_boot,
orig_sample_estimate - qnorm(p=0.025, mean=0, sd=1)*se_boot)
CI_95_se_method
```

```
0.5317028 1.1546139
```

```
qs_boot = quantile(boot_estimates, probs=c(0.025, 0.975))
CI_95_p_method = unname(qs_boot)
CI_95_p_method
```

```
0.603825 1.208100
```

```
# difference between medians
X1 = SignalsDF_2002$AvgMonRet[SignalsDF_2002$qNXF ==
1][!is.na(SignalsDF_2002$AvgMonRet[SignalsDF_2002$qNXF == 1])]
X2 = SignalsDF_2002$AvgMonRet[SignalsDF_2002$qNXF ==
5][!is.na(SignalsDF_2002$AvgMonRet[SignalsDF_2002$qNXF == 5])]
orig_sample1 = X1
orig_sample2 = X2
n1 = length(orig_sample1)
n2 = length(orig_sample2)

boot_samples1 = replicate(n=100000, sample(orig_sample1, n1, replace=TRUE),
simplify=FALSE)
boot_samples2 = replicate(n=100000, sample(orig_sample2, n2, replace=TRUE),
simplify=FALSE)
boot_estimates = sapply(boot_samples1, FUN=median, simplify=TRUE)-
sapply(boot_samples2, FUN=median, simplify=TRUE)
se_boot = sd(boot_estimates)
orig_sample_estimate = median(orig_sample1)-median(orig_sample2)

CI_95_se_method = c(orig_sample_estimate + qnorm(p=0.025, mean=0, sd=1)*se_boot,
orig_sample_estimate - qnorm(p=0.025, mean=0, sd=1)*se_boot)
CI_95_se_method
```

```
-0.9701848 -0.2645319
```

```
qs_boot = quantile(boot_estimates, probs=c(0.025, 0.975))
CI_95_p_method = unname(qs_boot)
CI_95_p_method
```

```
-1.0030750 -0.3149667
```

```
# Pearson correlation for qNXF = 1
orig_sample = SignalsDF_2002[c('AvgMonRet', 'Size')][SignalsDF_2002$qNXF == 1,]
n = dim(orig_sample)[1]
orig_index = 1:n

boot_estimates = rep(NA, times=100000)
for(i in 1:100000){
  boot_index = sample(orig_index, n, replace=TRUE)
  boot_sample = orig_sample[boot_index, ]
  boot_estimates[i] = cor(boot_sample[,1], boot_sample[,2], use="complete.obs",
method="pearson")
}

se_boot = sd(boot_estimates)
orig_sample_estimate = cor(orig_sample[,1], orig_sample[,2], use="complete.obs",
method="pearson")

CI_95_se_method = c(orig_sample_estimate + qnorm(p=0.025, mean=0, sd=1)*se_boot,
orig_sample_estimate - qnorm(p=0.025, mean=0, sd=1)*se_boot)
CI_95_se_method
```

```
-0.05764502 0.01416233
```

```
qs_boot = quantile(boot_estimates, probs=c(0.025, 0.975))
CI_95_p_method = unname(qs_boot)
CI_95_p_method
```

```
-0.05510957 0.01689215
```

```
# Pearson correlation for qNXF = 5
orig_sample = SignalsDF_2002[c('AvgMonRet', 'Size')][SignalsDF_2002$qNXF == 5,]
n = dim(orig_sample)[1]
orig_index = 1:n
```

```

boot_estimates = rep(NA, times=100000)
for(i in 1:100000){
  boot_index = sample(orig_index, n, replace=TRUE)
  boot_sample = orig_sample[boot_index, ]
  boot_estimates[i] = cor(boot_sample[,1], boot_sample[,2], use="complete.obs",
method="pearson")
}

se_boot = sd(boot_estimates)
orig_sample_estimate = cor(orig_sample[,1], orig_sample[,2], use="complete.obs",
method="pearson")

CI_95_se_method = c(orig_sample_estimate + qnorm(p=0.025, mean=0, sd=1)*se_boot,
                    orig_sample_estimate - qnorm(p=0.025, mean=0, sd=1)*se_boot)
CI_95_se_method

```

```

-0.081047947 -0.005719025

```

```

qs_boot = quantile(boot_estimates, probs=c(0.025, 0.975))
CI_95_p_method = unname(qs_boot)
CI_95_p_method

```

```

-0.09688543 -0.01941001

```

Since both lower and upper bound for the 95% CI of difference of medians is negative, we can say with 95% confidence that qNXF=5 portfolio has larger median than qNXF=1 portfolio.

Similarly, for qNXF = 5 portfolio, we can say that the Pearson correlation is negative with 95% confidence.

Also, there is some discrepancy in standard error method and percentile method for estimating CI, this is because standard error method assumes normal distribution. However, as we have found above, the distributions are not normal.

```

# vii
orig_sample = SignalsDF_2002$AvgMonRet[SignalsDF_2002$qNXF ==
1][!is.na(SignalsDF_2002$AvgMonRet[SignalsDF_2002$qNXF == 1])]
n = length(orig_sample)
obs_val = median(orig_sample)
obs_val
c = median(SignalsDF_2002$AvgMonRet[SignalsDF_2002$qNXF == 5], na.rm = TRUE)
trans_sample = orig_sample - obs_val + c
boot_samples = replicate(n=100000, sample(trans_sample, n, replace=TRUE),
simplify=FALSE)
boot_estimates = sapply(boot_samples, FUN=median, simplify=TRUE)

Boot_p_val = (sum(boot_estimates <= c - obs_val) + sum(boot_estimates >= c +
obs_val))/100000
Boot_p_val

```

```

0.01062

```

We reject the null hypothesis that medians are equal. This is indeed consistent with the inference drawn from the 95% CI for the difference of medians. Where we found that the median for qNXF = 5 distribution is larger than qNXF = 1 median with 95% confidence.

```

# viii
orig_sample1 = SignalsDF_2002$AvgMonRet[SignalsDF_2002$qNXF ==
1][!is.na(SignalsDF_2002$AvgMonRet[SignalsDF_2002$qNXF == 1])]
n1 = length(orig_sample1)
orig_sample2 = SignalsDF_2002$AvgMonRet[SignalsDF_2002$qNXF ==
5][!is.na(SignalsDF_2002$AvgMonRet[SignalsDF_2002$qNXF == 5])]
n2 = length(orig_sample2)

obs_val1 = skewness(orig_sample1)
obs_val2 = skewness(orig_sample2)
obs_val_stat = obs_val1 - obs_val2

```

```

orig_sample_comb = c(orig_sample1, orig_sample2)
boot_estimates = rep(NA, times=100000)
for(i in 1:100000){
  boot_sample_comb = sample(orig_sample_comb, n1 + n2, replace=TRUE)
  boot_sample1 = boot_sample_comb[1:n1]
  boot_sample2 = boot_sample_comb[(n1+1):(n1+n2)]
  boot_estimates[i] = skewness(boot_sample1) - skewness(boot_sample2)
}
c = 0
Boot_p_val = (sum(boot_estimates <= c + obs_val_stat) + sum(boot_estimates >= c -
obs_val_stat))/100000
Boot_p_val
0.87191

```

We fail to reject the null hypothesis that skewnesses are equal. The point estimates for skewnesses were 0.96 and 1.05.

Question 2.

Code:

```

#####
# q2
# i
SignalsDF_1992 = Signals[Signals$PFormDate == '1992-06-30',]

# ii
fit1 = lm(AvgMonRet ~ log(Size) + log(BTM) + ROE, data = SignalsDF_1992)
vcovHC0 = vcovHC(fit1, type = 'HC0')
coeftest(fit1, vcov. = vcovHC0)

```

t test of coefficients:

	Estimate	Std. Error	t value	Pr(> t)		
(Intercept)	4.067771	0.309085	13.1607	< 0.00000000000000022	***	
log(Size)	-0.293764	0.052694	-5.5749	0.00000002769114	***	
log(BTM)	0.640325	0.097001	6.6012	0.000000000005057	***	
ROE	-0.055731	0.121724	-0.4578	0.6471		

Signif. codes:	0 '***'	0.001 '**'	0.01 '*'	0.05 '.'	0.1 ' '	1

```
coefci(fit1, "log(Size)", level=0.95, vcov.=vcovHC0)
```

	2.5 %	97.5 %
log(Size)	-0.3970974	-0.1904316

```
coefci(fit1, "log(BTM)", level=0.95, vcov.=vcovHC0)
```

	2.5 %	97.5 %
log(BTM)	0.450105	0.8305444

```
coefci(fit1, "ROE", level=0.95, vcov.=vcovHC0)
```

	2.5 %	97.5 %
ROE	-0.2944313	0.1829702

Using the same approach as above, we can see that 95% CI's for the coefficients of $\log(\text{Size})$ and $\log(\text{BTM})$ do not contain zero. Therefore, there we can be 95% confident that those coefficient estimates are different from zero. T-statistics and p-values show exactly that, i.e. we reject the hypotheses that $\log(\text{Size})$ and $\log(\text{BTM})$ coefficient estimates are not significantly different from zero.

On the other hand, coefficient estimate for ROE has a 95% CI that contains zero, and as a result t-value is relatively small, and p-value is large. In this case, the confidence interval is narrow enough that we fail to reject the null hypothesis that coefficient estimate of ROE is significantly different from zero.

```
fit2 = felm(AvgMonRet ~ log(Size) + log(BTM) + ROE | 0 | 0 | BBHL,
data=SignalsDF_1992)
vcovCluster = fit2$clustervcov
coeftest(fit2,vcov.=vcovCluster)
```

t test of coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	4.067771	0.409379	9.9364	< 0.00000000000000022 ***
log(Size)	-0.293764	0.069069	-4.2532	0.00002193 ***
log(BTM)	0.640325	0.131002	4.8879	0.00000109 ***
ROE	-0.055731	0.130485	-0.4271	0.6693

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

```
coefci(fit2,"log(Size)",level=0.95,vcov.=vcovCluster)
```

	2.5 %	97.5 %
log(Size)	-0.4292093	-0.1583197

```
coefci(fit2,"log(BTM)",level=0.95,vcov.=vcovCluster)
```

	2.5 %	97.5 %
log(BTM)	0.3834297	0.8972197

```
coefci(fit2,"ROE",level=0.95,vcov.=vcovCluster)
```

	2.5 %	97.5 %
ROE	-0.3116127	0.2001516

The exact same interpretation applies here also. We fail to reject the null hypothesis that coefficient estimate for *ROE* is significantly different from zero.

```
# iii
jarque.bera.test(fit1$residuals)
```

Jarque Bera Test

data: fit1\$residuals
X-squared = 424536, df = 2, p-value < 0.00000000000000022

Jarque Bera test yields that the residuals of Equation 1 are not normally distributed. This means that OLS estimators are inconsistent (inefficient).

```
# iv
skewness(fit1$residuals)
```

4.482117

```
kurtosis(fit1$residuals)
```

69.23057

These values are consistent with result in (iii), because skewness is different from 0, and kurtosis is different from 3.

v) Since the OLS estimators are inconsistent, we cannot completely rely on the inferences we drew in part (ii).

```
# vi
orig_sample = SignalsDF_1992
n = dim(SignalsDF_1992)[1]
orig_index = 1:n
boot_estimatesBeta1 = rep(NA, times=10000)
boot_estimatesBeta2 = rep(NA, times=10000)
boot_estimatesBeta3 = rep(NA, times=10000)
for(i in 1:25000){
  boot_index = sample(orig_index, n, replace=TRUE)
  boot_sample = orig_sample[boot_index, ]
  fitx = lm(AvgMonRet ~ log(Size) + log(BTM) + ROE, data = boot_sample)
  boot_estimatesBeta1[i] = fitx$coefficients['log(Size)']
  boot_estimatesBeta2[i] = fitx$coefficients['log(BTM)']
  boot_estimatesBeta3[i] = fitx$coefficients['ROE']
}
### Using the standard error method
```

```

se_boot1 = sd(boot_estimatesBeta1)
se_boot2 = sd(boot_estimatesBeta2)
se_boot3 = sd(boot_estimatesBeta3)

orig_sample_estimate1 = lm(AvgMonRet ~ log(Size) + log(BTM) + ROE, data =
orig_sample)$coefficients['log(Size)']
orig_sample_estimate2 = lm(AvgMonRet ~ log(Size) + log(BTM) + ROE, data =
orig_sample)$coefficients['log(BTM)']
orig_sample_estimate3 = lm(AvgMonRet ~ log(Size) + log(BTM) + ROE, data =
orig_sample)$coefficients['ROE']

CI_95_se_method1 = c(orig_sample_estimate1 + qnorm(p=0.025, mean=0, sd=1)*se_boot1,
orig_sample_estimate1 - qnorm(p=0.025, mean=0, sd=1)*se_boot1)
# Beta1
CI_95_se_method1

```

log(Size)	log(Size)
-0.3970266	-0.1905024

```

CI_95_se_method2 = c(orig_sample_estimate2 + qnorm(p=0.025, mean=0, sd=1)*se_boot2,
orig_sample_estimate2 - qnorm(p=0.025, mean=0, sd=1)*se_boot2)
# Beta2
CI_95_se_method2

```

log(BTM)	log(BTM)
0.4483337	0.8323157

```

CI_95_se_method3 = c(orig_sample_estimate3 + qnorm(p=0.025, mean=0, sd=1)*se_boot3,
orig_sample_estimate3 - qnorm(p=0.025, mean=0, sd=1)*se_boot3)
# Beta3
CI_95_se_method3

```

ROE	ROE
-0.3838775	0.2724164

```

# Using the percentile method
qs_boot1 = quantile(boot_estimatesBeta1, probs=c(0.025, 0.975))
CI_95_p_method1 = unname(qs_boot1)
# Beta1
CI_95_p_method1

```

-0.4019474	-0.1950862
------------	------------

```

qs_boot2 = quantile(boot_estimatesBeta2, probs=c(0.025, 0.975))
CI_95_p_method2 = unname(qs_boot2)
# Beta2
CI_95_p_method2

```

0.4493491	0.8340377
-----------	-----------

```

qs_boot3 = quantile(boot_estimatesBeta3, probs=c(0.025, 0.975))
CI_95_p_method3 = unname(qs_boot3)
# Beta3
CI_95_p_method3

```

-0.4644347	0.2368468
------------	-----------

When comparing 95% CI's found via Bootstrap method vs the confidence intervals found via heteroskedasticity and cluster-robust variances, we can see that they are somewhat similar even though the distributions are not normal. Therefore, inferences found in part (ii) are still applicable.